PRACTICE TEST • MATHEMATICS PART 1

TESLA COACHING CLASSES

Std 10th SSC Maharashtra Board

PST-07

Subject: Mathematics Part 1 (Algebra)

Time: 2 Hours **Total Marks: 40**

SYLLABUS / CHAPTER WEIGHTAGE

- Chapter 1: Linear Equations in Two Variables — 30% (~12 marks)
- Chapter 2: Quadratic Equations — 30% (~12 marks)
- Chapter 3: Arithmetic Progression — 20% (~8 marks)
- Chapter 4: Financial Planning — 20% (~8 marks)

Q.1 (A) Choose the correct alternative for each of the following sub-questions: [4 Marks]

1. Find the value of D_x for the equations $4x + 3y = 19$ and $4x - 3y = -11$.

A. 24 B. -24
C. 44 D. 8

2. If the roots of the given quadratic equation are real and equal, then find the value of 'k':
$x^2 + kx + 1 = 0$.

A. ± 2 B. 4
C. -2 D. ± 4

3. First four terms of an A.P. are _____ whose first term is -2 and the common difference is -2.

A. $-2, 0, 2, 4$ B. $-2, -4, -6, -8$
C. $-2, 2, -2, 2$ D. $-2, -4, -8, -16$

4. In GST, all goods are classified by given numerical code called _____ code.

A. HSN B. SAC
C. GSTIN D. PAN

Q.1 (B) Solve the following sub-questions: [4 Marks]

1. If $D_x = 24$ and $x = -3$, then find the value of D.

2. If $a = 1, b = 4, c = -5$, then find the value of $b^2 - 4ac$.

3. Find a and d for an A.P., $1, 4, 7, 10, \ldots$

4. The rate of GST on stainless steel utensils is 18%, then the rate of State GST is _____.

Q.2 (A) Complete and write any TWO activities from the following: [4 Marks]

1. Complete the activity to find the value of determinant $\begin{vmatrix} 3 & 2 \\ 4 & 5 \end{vmatrix}$.

$\begin{vmatrix} 3 & 2 \\ 4 & 5 \end{vmatrix} = 3\times$ $\square$ $-2\times$ $\square$

$= \square - 8$

$= \square$

2. Complete the activity to find the value of discriminant for quadratic equation $x^2 + 10x - 7 = 0$.

Comparing $x^2 + 10x - 7 = 0$ with $ax^2 + bx + c = 0$,

$a = 1, b = 10, c = -7$

$b^2 - 4ac = (10)^2 - 4 \times 1\times$ $\square$

$= 100+$ $\square$

$= \square$

Discriminant $=$ $\square$

3. Complete the activity to find the sum of first 50 natural numbers.

First 50 natural numbers: $1, 2, 3, \ldots, 50$

Here $a = 1, d = 1, n = 50$

$S_n = \frac{n}{2}[_____] + (n-1)d$

$S_{50} = \frac{50}{2}[2(1) + (50-1)\times$ $\square$ $]$

$S_{50} = 25 \times [2+$ $\square$ $]$

$S_{50} = 25 \times 51 =$ $\square$

Q.2 (B) Solve any FOUR sub-questions from the following: [8 Marks]

1. If $x + 2y = 5$ and $2x + y = 7$, then find the value of $x + y$.

2. Solve the following quadratic equation by factorization method. $2x^2 - 2x + \frac{1}{2} = 0$

3. Find the 18th term of the A.P. $1, 7, 13, 19, \ldots$

4. Malik Gas Agency supplied LPG cylinder to a consumer for taxable value of $\text{Rs. } 545$. GST charged is 5% . Find the amount of CGST and SGST to be paid by the consumer.

5. Solve the quadratic equation by formula method. $x^2 + 10x + 2 = 0$

Q.3 (A) Complete and write any ONE activity from the following: [3 Marks]

1. Complete the activity to solve the simultaneous equations using Cramer's rule: $y + 2x - 19 = 0$ and $2x - 3y + 3 = 0$.

Writing in standard form: $2x + y = 19$ and $2x - 3y = -3$

$D = \begin{vmatrix} 2 & 1 \\ 2 & -3 \end{vmatrix} =$ ☐

$D_x = \begin{vmatrix} 19 & 1 \\ -3 & -3 \end{vmatrix} =$ ☐

$D_y = \begin{vmatrix} 2 & 19 \\ 2 & -3 \end{vmatrix} =$ ☐

$x = \frac{D_x}{D} =$ ☐

$y = \frac{D_y}{D} =$ ☐

Solution $(x, y) =$ (_____, _____)

2. Complete the activity to find the number of terms in an A.P. $12, 16, 20, \ldots, 132$.

Given A.P. is $12, 16, 20, \ldots, 132$

Here $a = 12, d =$ ☐, $t_n = 132$

$t_n = a + (n-1)d$

$132 = 12 + (n-1) \times$ ☐

$120 = (n-1) \times$ ☐

$n - 1 =$ ☐

$n =$ ☐

There are ☐ terms in the given A.P.

Q.3 (B) Solve any TWO sub-questions from the following: [6 Marks]

1. Solve the following simultaneous equations using Cramer's rule: $4x + 3y = 24$ and $3x + 4y = 25$.

2. Mukund has $\text{Rs. } 50$ more than Sagar. If the product of the amount they have is $15,000$, find the amount each one has.

3. Find the sum of all even natural numbers between 1 and 350.

4. Market value of a share is $\text{Rs. } 200$. If the brokerage rate is 0.3%, then find the purchase value of the share.

Q.4 Solve any TWO sub-questions from the following: [8 Marks]

1. A person starts a job with fixed salary and yearly increment. Salary after 4 years is $\text{Rs. } 15,000$ and after 10 years is $\text{Rs. } 18,000$. Find his initial salary and yearly increment.

2. A tank fills completely in 2 hours if both taps are opened. If only one tap is open, the smaller tap takes 3 hours more than the larger tap to fill the tank. How much time does each tap take to fill the tank completely?

3. On New Year's day Sanjana decides to save $\text{Rs. } 100$ on the first day, $\text{Rs. } 120$ on the second day, $\text{Rs. } 140$ on the third day and so on. What will be her total savings in 30 days?

Q.5 Solve any ONE of the following sub-questions: **[3 Marks]**

1. Construct any one meaningful word problem on simultaneous equations in two variables where the answer should be 5 and 3.

2. Form a quadratic equation such that one of its roots is $2 + \sqrt{5}$.
